

Finding Nicole's home: a data-driven shortlist

King County housing data · May 2014 – May 2015

One year of sales, 70 zip codes — enough to compare neighbourhoods

- 21,597 sales of 21,420 homes, May 2014 – May 2015 — 177 homes sold twice (real re-sales, kept)
- 70 zip codes across King County, 50–602 sales each (median 283)
- 21 columns per sale — price, size, rooms, build quality, location, plus the 15 nearest neighbours
- The market it covers: \$78K–\$7.7M (median \$450K), 370–13,540 sqft, built 1900–2015
- Gaps are small and fillable — yr_renovated 17.8%, waterfront 11.1%, sqft_basement 2.1%, view 0.3%: filled in, no rows thrown away
- Not in the data: walkability, nightlife, commute time — so "lively" needs a proxy

WHAT'S AVAILABLE — 21 COLUMNS PER SALE

Price & date

price, date

Size

sqft_living, sqft_lot, sqft_above, sqft_basement

Rooms

bedrooms, bathrooms, floors

Build quality

grade, condition, view, waterfront, yr_built, yr_renovated

Location

zipcode, lat, long

Neighbours (15 nearest)

sqft_living15, sqft_lot15

Meet Nicole — a single first-time buyer, no kids

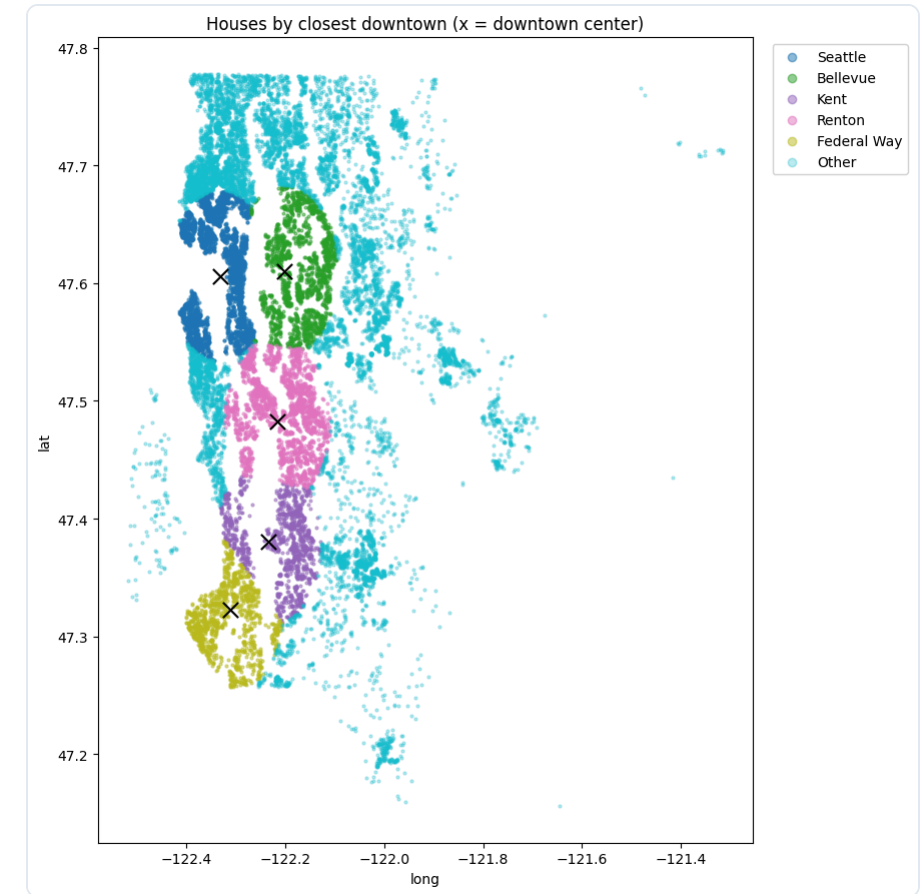
- 1–2 bedrooms, 485–645 sqft
- Wants lively and central, on a mid-range budget
- Buying within the next 12 months

HOW WE DEFINED "LIVELY AND CENTRAL"

- **Central** = within 5 miles of one of the 5 biggest cities — Seattle, Bellevue, Kent, Renton, Federal Way
- **Lively** = dense, using average lot size of the 15 nearest neighbours as the proxy

WHAT WE WANT TO CONFIRM OR REJECT

- Downtown sale prices are **50%+ higher** than the outskirts
- Downtowns are **denser** than the outskirts
- **Winter** sale prices are lower (up to 5%)



Only 2 of 5 downtowns are actually pricier than the outskirts

THE ASSUMPTION THIS RESTS ON

Downtown = within 5 miles of a city centre. Straight-line (haversine) distance from each home to the nearest of the 5 city centres; beyond 5 miles from all of them, it's outskirts. Not drive time, and the 5-mile radius is our choice, not the data's.

- Pooling all 5 downtowns hid the effect (not significant, $p = 0.46$)
- Tested individually, each city vs. its own local outskirts:
 - **Bellevue is 31.9% pricier** than its own outskirts (significant, but under the 50% bar)
 - **Seattle is 29.1% pricier** (significant, but under the 50% bar)
 - Renton (-15.4%), Kent (-9.0%), Federal Way (-12.7%) — cheaper than their own outskirts, not significant

MEDIAN PRICE: DOWNTOWN VS. THAT CITY'S OWN OUTSKIRTS

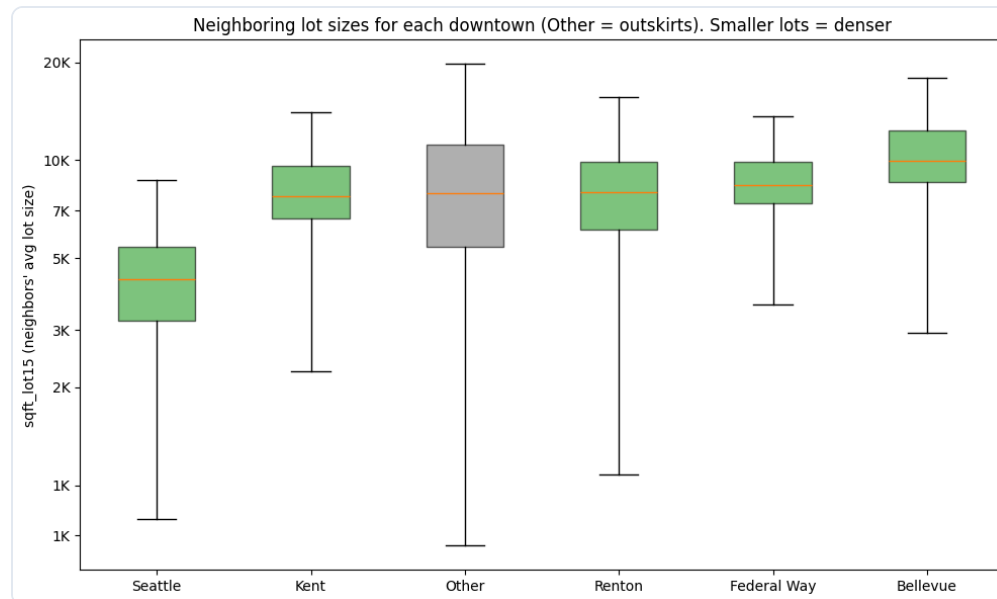
TOWN	DOWNTOWN MEDIAN	OUTSKIRTS MEDIAN	% DIFF
Bellevue	\$745,000	\$565,000	+31.9%
Seattle	\$569,950	\$441,375	+29.1%
Renton	\$330,000	\$390,000	-15.4%
Kent	\$286,543	\$314,950	-9.0%
Federal Way	\$262,250	\$300,250	-12.7%

Seattle is the only downtown that's genuinely denser than the outskirts

THE ASSUMPTION THIS RESTS ON

Smaller average lot size = denser = livelier. sqft_lot15, the average lot size of a home's 15 nearest neighbours, stands in for density — and density stands in for "lively." Nothing in the data measures walkability or nightlife, so it's a proxy, not a measurement.

- Downtown overall is 7.7% denser (significant)
- By city:
 - **Seattle's lots are 45.4% smaller** than outskirts (significant)
 - Kent, Renton, Federal Way — no significant difference
 - Bellevue — actually 26% *bigger*, not smaller



11 homes match Nicole's criteria today — all in Seattle

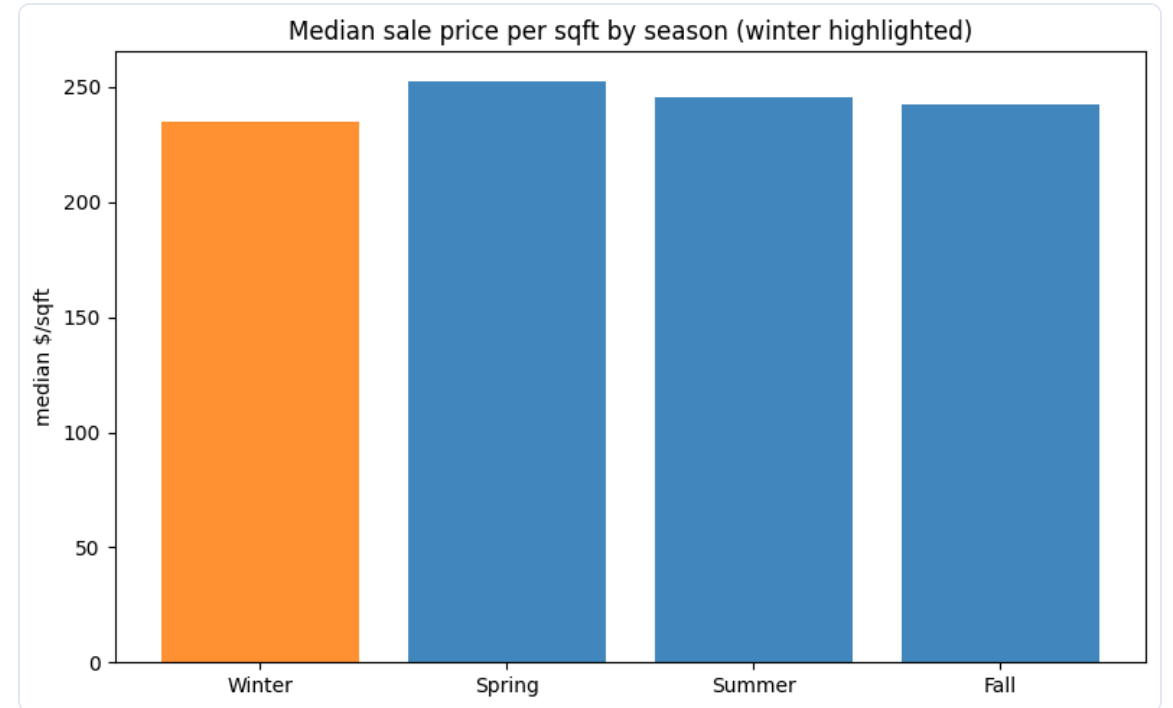
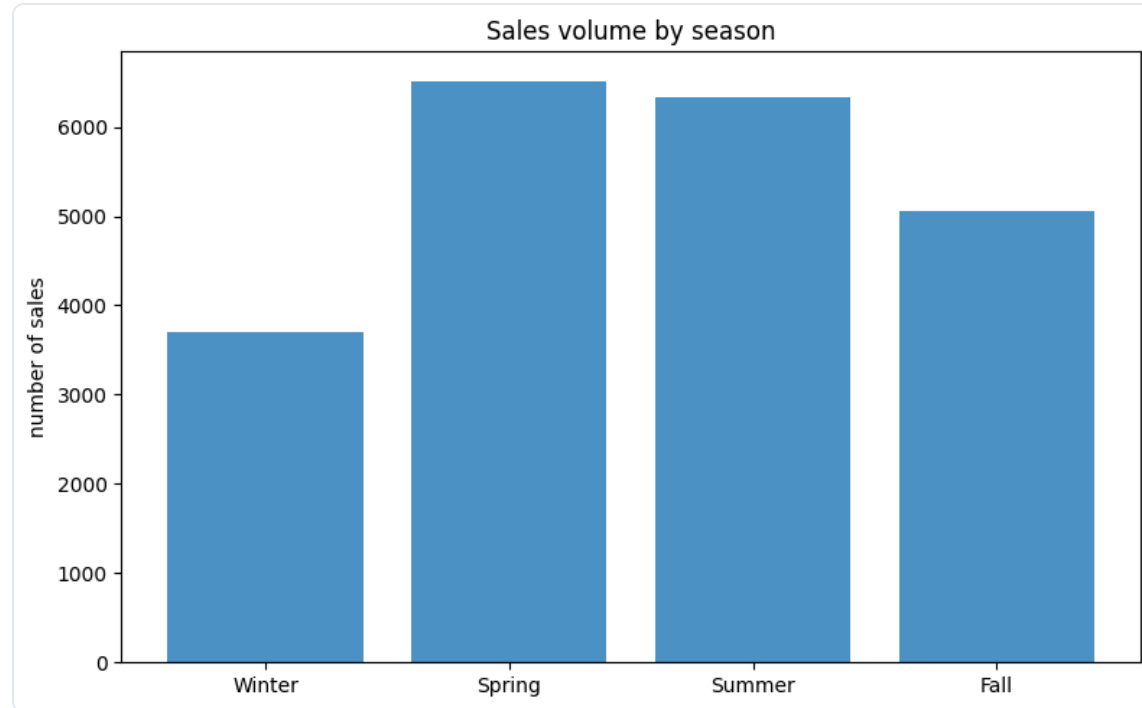
- Her size range (485–645 sqft, 1–2BR) in Seattle: \$280,500–\$383,944 (middle 50% of the market)
- Capping at that price ceiling: 11 matching homes, all in Seattle
- Ranked by build quality (grade, then condition)

11-HOME SHORTLIST, RANKED BY GRADE/CONDITION

GRADE	COND.	BEDS	SQFT	PRICE
7	4	1	590	\$202,000
7	3	1	550	\$353,000
6	4	2	610	\$286,000
6	3	1	520	\$295,000
6	3	1	620	\$382,888
6	3	1	600	\$229,000
6	3	1	640	\$340,000
5	3	2	630	\$315,000
5	3	1	520	\$275,000
5	2	1	570	\$310,000
5	2	2	590	\$156,000

Spring is when the market peaks — but the **winter discount** is real

- Sales aren't spread evenly across the year ($p \approx 0$)
- Peak season: **Spring (30.2%)**, not Summer (29.3%) as assumed
- Winter homes sold for **5.5% less**
- Checked per sqft too: still **4.9% cheaper** than the rest of the year
- Spring is also the priciest season per sqft — winter runs **7.1% below** it



Three takeaways for Nicole

- Focus the search on **Seattle** — the only downtown that's genuinely lively (dense); budget roughly **\$280K–\$384K**
- Start with the **11 matching homes**, best-graded and best-condition first
- **Buy in winter** — the discount holds even after controlling for house size, worth roughly **\$12K–\$27K** against buying at the spring peak

HOW "SAVINGS VS. SPRING" IS WORKED OUT

1. Each season's median \$/sqft ÷ the all-year median gives it an index: **Winter 0.959**, Spring 1.033, Summer 1.004, Fall 0.990.
2. Divide a home's price by its own sale season's index, multiply by the target season's.
3. Winter vs. spring, the priciest season: a **7.1% gap** on every home. Uses \$/sqft, so house size doesn't skew it.

FINAL SHORTLIST, PRICED FOR WINTER

GRADE	COND.	BUILT	BEDS	SQFT	WINTER PRICE	SAVINGS VS. SPRING
7	4	1926	1	590	\$193,071	\$14,823
7	3	2008	1	550	\$327,831	\$25,169
6	4	1918	2	610	\$273,358	\$20,987
6	3	1918	1	520	\$285,815	\$21,943
6	3	1900	1	620	\$355,588	\$27,300
6	3	1920	1	600	\$212,672	\$16,328
6	3	1918	1	640	\$329,414	\$25,291
5	3	1918	2	630	\$292,540	\$22,460
5	3	1930	1	520	\$275,000	\$21,113
5	2	1918	1	570	\$300,348	\$23,059
5	2	1947	2	590	\$151,143	\$11,604